

REG NO: 22TD0053 NAME: MOHAMED B SIRAJUDDEEN

YEAR/SEM/SEC: Y3/S6/B

MINI PROJECT

HOTEL MANAGEMENT SYSTEM

DESCRIPTION:

A Hotel Management System (HMS) is a comprehensive software solution designed to streamline and automate the operations of a hotel or hospitality business. It serves as a central platform to manage various aspects of hotel operations, from reservations and guest services to financial transactions and reporting..A Hotel Management System is essential for efficient and organized hotel operations. It helps improve guest experiences, streamline internal processes, and drive revenue. The specific features and capabilities can vary depending on the software provider and the needs of the hotel or hospitality business.

REQUIREMENT ANALYSIS:

Preliminary Requirements Analysis:

1) Database:

- i) To store reservation detail.
- ii) To store guest detail.
- iii) To store payment detail.
- iv) To store rooms information.

2) Web Interface:

- i) Allow the reception to modify, add or delete reservation.
- ii) Allow the reception to modify, add or delete guest detail.
- iii). Allow the reception to provide the guest with the payment detail.

PROCEDURE:

STEP 1: Database Design: Design a database to store information about rooms, guests, reservations, and other relevant data.

STEP 2: User Interface: Create a user-friendly interface with Windows Forms, including forms for room booking, guest check-in, and check-out.

STEP 3: Implement features to manage rooms, such as adding, editing, and deleting room details.

STEP 4: Develop functionality to manage guest information, including check-in and check-out processes.

STEP 5: Create a reservation system to allow guests to book rooms in advance.

STEP 6: Implement billing and payment features, including generating invoices and processing payments.

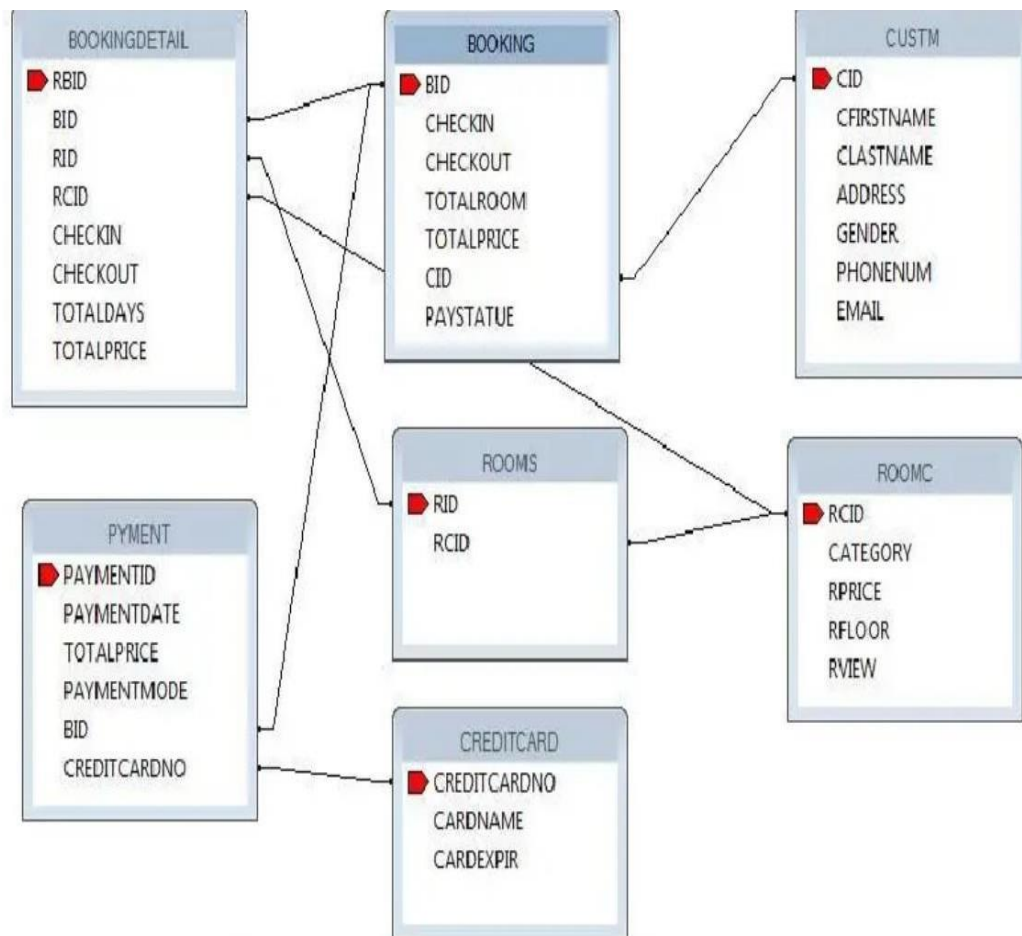
STEP 7: Develop reporting features for generating various reports like occupancy, revenue, and guest lists.

STEP 8: Ensure data security by implementing user authentication and authorization to restrict access to certain features.

STEP 9: Validate user input to prevent errors and ensure data integrity.

STEP 10: Implement error handling mechanisms to gracefully handle unexpected issues and provide informative error messages to users.

DATA FLOW DIAGRAM (DFD):



SAMPLE CODING:

THE CODE FOR THE TEXT BOX:

```
<asp:TextBox ID="TextBox5" runat="server">
```

CONNECTION BETWEEN THE WEB AND THE DATABASE:

```
Dim Conn As New OleDbConnection( "Provider=OraOLEDB.Oracle.1;Data  
Source=localhost:UserID=HR:Password=amanigeama;Unicode=True")
```

INSERTION FROM THE WEB INTO DATABASE:

CheckIn Date *		Customer Details
CheckOut Date *		
Room 1 *	*	First Name *
Room 2		Last Name *
Room 3		Address
Room 4		E-mail *
Room 5		Tel.Number
Room 6		ID Number *

```
Dim Conn As New OleDbConnection( "Provider=OraOLEDB.Oracle.1;Data
    Source=localhost; UserID=HR:Password-
    amanigeama;Unicode=True")

Dim MyCmd As New OleDbCommand

Conn.Open()

Dim InsertCommand As New OleDbCommand

Dim CMD As New OleDbCommand

CMD.Connection = Conn

CMD.CommandType = CommandType.Text

CMD.CommandText = "SELECT RCID FROM RoomC WHERE RCID = '" &
    TextBox2.Text & "'"

Dim reader As OleDbDataReader

reader = CMD.ExecuteReader

If reader.Read Then

InsertCommand.Connection = Conn

InsertCommand.CommandType = CommandType.Text

InsertCommand.CommandText = "INSERT INTO RoomS(RID,RCID)
    VALUES(?,?)"
```

```

InsertCommand.Parameters.Clear()
InsertCommand.Parameters.AddWithValue( "?". TextBox1.Text)
InsertCommand.Parameters.AddWithValue("?". TextBox2.Text)
InsertCommand.ExecuteNonQuery()
InsertCommand.Dispose()
Conn.Close()
MsgBox("OK". MsgBoxStyle.Information)
Elseif Not reader.Read Then

MsgBox(" The room category not exist in Room Category table",
MsgBoxStyle.Information)

End If

```

DELETING FROM THE DATABASE BY USING THE WEB:

The screenshot shows a web application interface for managing bookings. It is divided into two main panels. The left panel, titled "Booking Detail", contains several input fields and controls: "Check in" and "Check out" text boxes, "View" and "Type of Room" dropdown menus (both currently set to "All"), and a "Search" button. The right panel contains a "BookingNo" text box and three buttons labeled "Detail", "Modify", and "Pay".

```

Dim Conn As New OleDbConnection( "Provider=OraOLEDB.Oracle. 1;Data
    Source=localhost:UserID=HR:Password=amanigeama:Unicode=True")

Conn.Open()

Dim DeleteCommand As New OleDbCommand

DeleteCommand.Connection = Conn

Dim CMD As New OleDbCommand

CMD.Connection = Conn

CMD.CommandType = CommandType.Text

```

```

CMD.CommandText = "SELECT RID FROM RoomS WHERE RID='" &
    TextBox1.Text &'"

Dim reader As OleDbDataReader
reader = CMD.ExecuteReader

If reader.Read Then

DeleteCommand.CommandType = CommandType.Text

DeleteCommand.CommandText = "DELETE FROM RoomS WHERE RID=?"

DeleteCommand.Parameters.Clear()

DeleteCommand.Parameters.AddWithValue( "?", TextBox1.Text)

DeleteCommand.ExecuteNonQuery()

DeleteCommand.Dispose()

End If

```

UPDATING THE DATA:

The screenshot shows a web application interface for managing bookings. It is divided into two main panels. The left panel, titled "Booking Detail", contains several input fields and controls: "Check in" and "Check out" text boxes, "View" and "Type of Room" dropdown menus (both currently set to "All"), and a "Search" button. The right panel contains a "BookingNo" text box and three buttons labeled "Detail", "Modify", and "Pay".

```

Dim Conn As New OleDbConnection( "Provider=OraOLEDB.Oracle. 1;Data
    Source=localhost;UserID=HR:Password=amanigeama:Unicode=True")

Conn.Open()

Dim UpdateCommand As New OleDbCommand

UpdateCommand.Connection = Conn

Dim CMD As New OleDbCommand

CMD.Connection = Conn

```

```
CMD.CommandType = CommandType.Text

CMD.CommandText = "SELECT RCID FROM RoomC WHERE RCID = " &
    TextBox2.Text & ""

Dim reader As OleDbDataReader
reader = CMD.ExecuteReader

If reader.Read Then

    UpdateCommand.CommandType = CommandType.Text
    UpdateCommand.CommandText = "UPDATE RoomS set RCID=? WHERE
        RID =?"

    UpdateCommand.Parameters.Clear()
    UpdateCommand.Parameters.AddWithValue( "?", TextBox2.Text)
    UpdateCommand.Parameters.AddWithValue( "?", TextBox1.Text)
    UpdateCommand.ExecuteNonQuery()
    UpdateCommand.Dispose()

    Conn.Close()

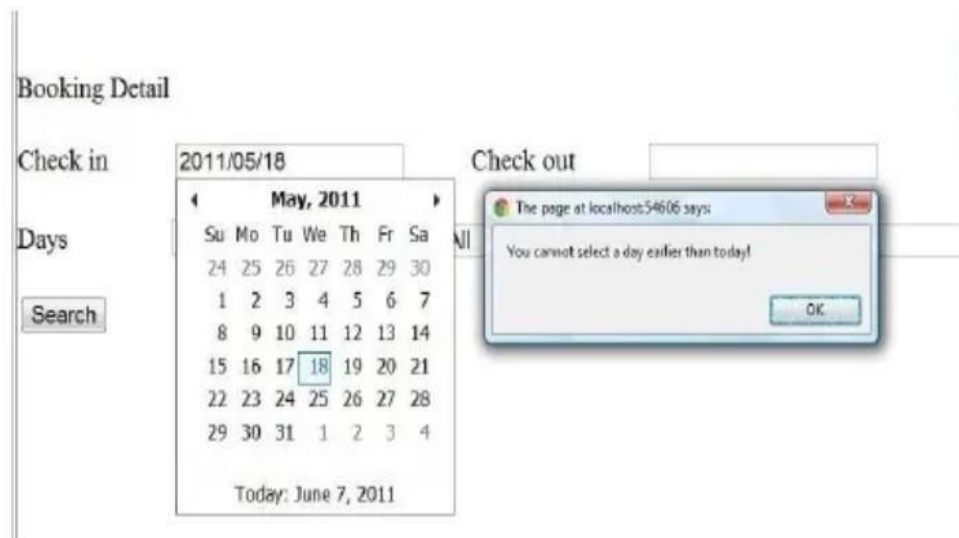
    MsgBox(" OK", MsgBoxStyle.Information)

Elseif Not reader.Read Then

    MsgBox(" The room category does not exist in Room Category table ".
        MsgBoxStyle.Information)

End If
```

ADDING THE CALENDAR EXTENSION:



For Adding calendar extender, the toolkitScriptManager has to be added. Then the calendar is dragged from the Ajax toolbox:

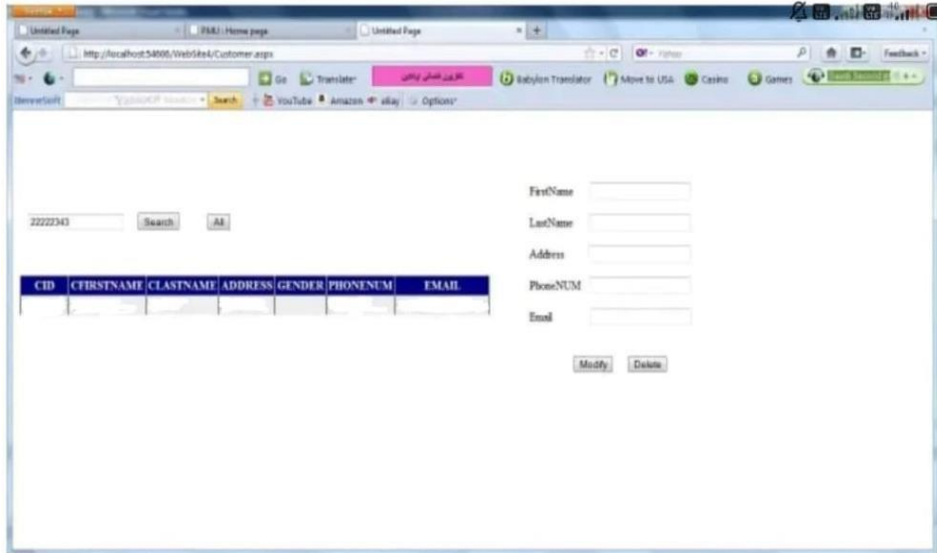
```
<asp:ToolkitScriptManager ID="ToolkitScriptManager1" runat="server">
</asp:ToolkitScriptManager>
<asp:TextBox ID="txtStartDate" runat="server"></asp:TextBox>
<asp:CalendarExtender ID="CalendarExtender1" TargetControlID="txtStartDate"
    runat="server" Format="yyyy/MM/dd" OnClientDateSelectionChanged =
    "checkDate" />
```

TRANSFER VALUE FROM PAGE TO ANOTHER PAGE:

```
Session("field88") = txtStartDate.Text
```

```
Session("field99") = TextEndDate.Text
```


SEARCH CODE:



```
Dim Conn As New OleDbConnection("Provider=OraOLEDB.Oracle.1;Data  
Source=localhost;  
UserID=HR;Password=amanigeama;Unicode=True")
```

```
Dim MyCmd As New OleDbCommand
```

```
Dim myDataSet As New DataSet
```

```
myDataSet.Clear()
```

```
Conn.Open()
```

```
MyCmd.Connection = Conn
```

```
MyCmd.CommandType = CommandType.Text
```

```
Dim currentDate As Date
```

```
currentDate = System.DateTime.Now
```

```
If txtStartDate.Text < TextEndDate.Text Then
```

```
If System.DateTime.Now < txtStartDate.Text Or System.DateTime.Now <  
TextEndDate.Text Then
```

```
If DropDownList2.Items(0).Selected = True Then
```

```
If DropDownList1.Items(0).Selected = True Then
```

```
MyCmd.CommandText = "Select RS.rid, RC.RCID, RC.Category, RC.Rprice,  
RC.RFloor, RC.Rview FROM RoomC RC, RoomS RS WHERE RC.RCID =  
RS.RCID AND RS.rid NOT IN (SELECT RS.rid FROM RoomC
```

```
RC, BOOKINGDETAIL BD WHERE (" & txtStartDate.Text & " <= BD.CheckIn  
AND " & TextEndDate.Text & ">=BD. CheckOut )AND RS.rid=BD.rid)"
```

```
Elseif DropDownList1.Items(2).Selected = True Then
```

```
MyCmd.CommandText = "Select RS.rid, RC.RCID, RC.Category, RC.Rprice,  
RC.RFloor, RC.Rview FROM RoomC RC, RoomS RS WHERE RC.RCID =  
RS.RCID AND RC.category=" & DropDownList1.Items(2).Value & " AND RS.rid  
NOT IN (SELECT RS.rid FROM RoomC RC, BOOKINGDETAIL BD WHERE (" &  
& txtStartDate.Text & " <= BD.CheckIn AND " & TextEndDate.Text & "  
>=BD.CheckOut) AND RS.rid = BD.rid)"
```

```
Elseif DropDownList1.Items(3).Selected= True Then
```

```
MyCmd.CommandText = "Select RS.rid, RC.RCID, RC.Category, RC.Rprice,  
RC.RFloor, RC.Rview FROM RoomC RC, RoomS RS WHERE RC.RCID =  
RS.RCID AND RC.category=" & DropDownList1.Items(3).Value & " AND RS.rid  
NOT IN (SELECT RS.rid FROM RoomC RC, BOOKINGDETAIL BD WHERE (" &  
& txtStartDate.Text & "...
```

```
End If
```

```
Elseif DropDownList2.Items(1).Selected= True Then
```

```
If DropDownList1.Items(0).Selected = True Then
```

```
MyCmd.CommandText = "Select RS.rid, RC.RCID, RC.Category, RC.Rprice,  
RC.RFloor, RC.Rview FROM RoomC RC, RoomS RS WHERE RC.RCID =  
RS.RCID AND RC.Rview=" & DropDownList2.Items(1).Value & " AND RS.rid  
NOT IN (SELECT RS.rid FROM RoomC RC, BOOKINGDETAIL BD WHERE (" &  
& txtStartDate. Text & " <= BD.CheckIn AND " & TextEndDate.Text & ">=  
BD.CheckOut) AND RS.rid=BD.rid)"
```

```
Elseif DropDownList1.Items(1).Selected = True Then
```

```
MyCmd.CommandText = "Select RS.rid, RC.RCID, RC.Category, RC.Rprice,  
RC.RFloor, RC.Rview FROM RoomC RC, RoomS RS WHERE RC.RCID =  
RS.RCID AND RC.Rview=" & DropDownList2.Items(1).Value & "AND  
RC.category=" & DropDownList 1.Items(1).Value & "...
```

```
NOT IN (SELECT RS.rid FROM RoomC RC, BOOKINGDETAIL BD WHERE (&  
txtStartDate.Text & " <= BD.CheckIn AND " & TextEndDate.Text & ">=  
BD.CheckOut) AND RS.rid= BD.rid)"
```

Elseif DropDownList1.Items(4).Selected = True Then

```
MyCmd.CommandText = "Select RS.rid, RC.RCID, RC.Category, RC.Rprice,
RC.RFloor, RC.Rview FROM RoomC RC, RoomS RS WHERE RC.RCID =
RS.RCID AND RC.Rview=""" & DropDownList2.Items(1).Value & """ AND
RC.category=""" & DropDownList1.Items(4).Value & """ AND RS.rid NOT IN
(SELECT RS.rid FROM RoomC RC, BOOKINGDETAIL BD WHERE ("" &
txtStartDate.Text & "" <= BD.CheckIn AND "" & TextEndDate.Text & ">=
BD.CheckOut) AND RS.rid = BD.rid)"
```

Elseif DropDownList1.Items(5).Selected = True Then

```
MyCmd.CommandText = "Select RS.rid, RC.RCID, RC.Category, RC.Rprice,
RC.RFloor, RC.Rview FROM RoomC RC, RoomS RS WHERE RC.RCID =
RS.RCID AND RC.Rview = "" & DropDownList2.Items(2).Value & "AND
RC.category=" & DropDownList1.Items(1).Value & "" AND RS.rid NOT IN
(SELECT RS.rid FROM RoomC RC, BOOKINGDETAIL BD WHERE ("" &
txtStartDate.Text & " <= BD.CheckIn AN D^ prime prime prime &
TextEndDate.Text & ">= BD.CheckOut) AND RS.rid = BD.rid)"
```

Elseif DropDownList1.Items(2).Selected = True Then

```
MyCmd.CommandText = "Select RS.rid, RC.RCID, RC.Category, RC.Rprice,
RC.RFloor, RC.Rview FROM RoomC RC, RoomS RS WHERE RC.RCID =
RS.RCID AND RC.Rview = " & DropDownList2.Items(2).Value & " AND
RC.category=" & DropDownList1.Items(5).Value & "" AND RS.rid NOT IN
(SELECT RS.rid FROM RoomC RC, BOOKINGDETAIL BD WHERE (&
txtStartDate.Text & " <= BD.CheckIn AN D^ prime prime prime TextEndDate.Text
& ">= BD.CheckOut) AND RS.rid = BD.rid)"
```

End If

End

If

```
MyCmd.Parameters.Clear()
```

```
MyCmd.Parameters.AddWithValue( "?", txtStartDate.Text)
```

```
MyCmd.Parameters.AddWithValue( "?". TextEndDate.Text)
```

```
Dim adapter As New OleDb.OleDbDataAdapter(MyCmd.CommandText, Conn)
adapter.Fill(myDataSet. "0")
```

```
GridView1.DataSource = myDataSet.Tables("0")
```

```
GridView1.DataBind()
```

Else

```

MsgBox(" you have chose a previous data")
End If
Else
MsgBox(" the checkin data is bigger than the check out date")
End If
Conn.Close()
LOGIN CODE:

```

```

<LoggedInTemplate>
<span class="input">You are logged in. Welcome </span>
<asp:LoginName ID="LoginName1" runat="server" />
</LoggedInTemplate>
<AnonymousTemplate>
You are not logged in. Click the Login link to sign in.
<asp:Button ID="button7" runat="server" onclick="button7_Click"
Text="Reminder" />

```

</AnonymousTemplate>

</asp:LoginView>

<asp:LoginStatus ID="LoginStatus 1" runat="server" />

CHECKING EXIT OF THE CUSTOMER ID CODE:

Using reader to check if the customer ID is exit or not, before doing insert to the Guest detail.

```
Dim CMD As New OleDbCommand
```

```
CMD.Connection = Conn
```

```
CMD. Command * dType = CommandType.Text
```

```
CMD. Command * dText ="SELEC rceil CID FROM CUSTM WHERE CID = "" &  
TextBox7.Text &
```

```
Dim reader As OleDbDataReader
```

```
reader = CMD.ExecuteReader
```

```
'INSERT INTO THE CUSTOMER TABLE
```

```
If Not reader.Read Then
```

```
'INSERT INTO CUTOMER TABLE
```

```
Dim InsertCUST As New OleDbCommand
```

```
InsertCUST.Connection = Conn
```

```
InsertCUST.CommandType = CommandType.Text
```

```
InsertCUST.CommandText = "INSERT INTO  
CUSTM(CID,CFirstName,CLastName, Address, PhoneNUM,Email)
```

```
VALUES(?,?,?,?,?)"
```

```
InsertCUST.Parameters.Clear()
```

```
InsertCUST.Parameters.AddWithValue( "?" . TextBox7.Text)
```

```
InsertCUST.Parameters.AddWithValue( "?" . TextBox2.Text)
```

```
InsertCUST.Parameters.AddWithValue( "?" . TextBox3.Text)
```

```
InsertCUST.Parameters.AddWithValue( "?" . TextBox4.Text)
```

```
InsertCUST.Parameters.AddWithValue( "?". TextBox6.Text)
```

```
InsertCUST.Parameters.AddWithValue( "?". TextBox5.Text)
```

```
InsertCUST.ExecuteNonQuery()
```

```
InsertCUST.Dispose()
```

```
End If
```

```
reader.Close()
```

```
MsgBox(" OK", MsgBoxStyle. Information)
```

CALCULATING THE TOTAL COST:

```
Dim day As Int32
```

```
Dim chi As Date
```

```
Dim cho As Date
```

```
Dim price As Int16
```

```
price = 0
```

```
chi = CheckIn.Text
```

```
cho = Checkout.Text
```

```
day = Date Diff(DateInterval.Day, chi, cho)
```

```
Dim RomPrice As New OleDbCommand
```

```
RomPrice.Connection = Conn
```

```
RomPrice.CommandType = CommandType.Text
```

```
RomPrice.CommandText = "SELECT Rprice from RoomC WHERE RCID="" &  
TextBox 19.Text & ""
```

```
Dim Room As New OleDb.OleDbDataAdapter(RomPrice.CommandText, Conn)
```

```
Room.Fill(myDataSet. "RI")
```

```
price= myDataSet .Tables(^ prime prime R1^ prime prime ).Rows(0).Item(^ prime  
prime Rprice^ prime prime )
```

```
price = price* day
```

```
myDataSet.Clear()
```

OUTPUT:

RESERVATION FORM:

CheckIn Date *	2011/06/14	Customer Details		
CheckOut Date *	2011/06/30	First Name *	Fatimah	
Room 1 *	R160	RC11	Last Name *	Hadi
Room 2	R160	RC13	Address	Awamia
Room 3			E-mail *	200700818@pmu.edu.sa
Room 4			Tel.Number	
Room 5			ID Number *	200700818
Room 6				

CHECKING INSERTED DATE:

Booking Detail			
Check in	2011/05/18	Check out	
Days	Nil		
Search			

May, 2011

Su	Mo	Tu	We	Th	Fr	Sa
24	25	26	27	28	29	30
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31	1	2	3	4

Today: June 7, 2011

The page at localhost:54606 says:

You cannot select a day earlier than today!

OK

SEARCH RESULT:

Days: 1 View: PMU Type of Room: Deluxe Room

Button

RID	CATEGORY	RPRICE	RFLOOR	RVIEW
☐ R280	Deluxe Room	140	Secon Floor	PMU
☐ R290	Deluxe Room	140	Secon Floor	PMU
☒ R331	Deluxe Room	150	Third Floor	PMU
☐ R390	Deluxe Room	150	Third Floor	PMU
☒ R410	Deluxe Room	150	Fourth Floor	PMU
☐ R490	Deluxe Room	150	Fourth Floor	PMU
☐ R590	Deluxe Room	160	Fifth Floor	PMU
☐ R515	Deluxe Room	160	Fifth Floor	PMU

Button

Done Local intranet | Protected Mode Off 125%

ADDING CALENDAR:

Booking Detail

Check in: [] Check out: []

Days: [] All []

Search

Calendar: June, 2011

Su	Mo	Tu	We	Th	Fr	Sa
29	30	31	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	1	2
3	4	5	6	7	8	9

Today: June 7, 2011

CREDIT CARD FORM:

Credit Card

Credit Card Number *

200700818

Credit Card Holder Name *

Fatimah

Expire Date

01

2011

CALCULATING TOTAL COST:

Payment

Booking No

699994

PaymentDate

2011/06/22

Cash

CreditcardNo

200700818

Total Cost

555

CFIRSTNAME	CLASTNAME	ADDRESS	PHONENUM	EMAIL	CID
Fatimah	Yousif			200700818@pmu.edu.sa	122227777

BID	CHECKIN	CHECKOUT	TOTALROOM	CID	CONFIRMSTATUE
700000	23/06/2011 12:00:00 AM	25/06/2011 12:00:00 AM		122227777	Not confirmed

RBID	BID	RID	RCID	CHECKIN
5005	700000	R140	RC14	23/06/2011 12:00:00 AM
5006	700000	R170	RC11	23/06/2011 12:00:00 AM
5007	700000	R230	RC23	23/06/2011 12:00:00 AM

SEARCHING THE DETAILS OF THE GUEST:

CID	CFIRSTNAME	CLASTNAME	ADDRESS	GENDER	PHONENUM	EMAIL
20110111	Ahmed	Naser	Dammam	M	038550645	anaser@hotmail.com
20110112	Ayoub	Zaher	Dammam	M	038530645	azaher@hotmail.com
20110113	Hani	Nemer	Dammam	M	038550635	hnemer@hotmail.com
20110114	Eva	David	Dammam	F	018550655	edavid@hotmail.com
20110115	Krestin	MM	Dammam	F	028550665	kmm@hotmail.com
20110116	Mohammed	Nader	Dammam	M	028550675	mnader@hotmail.com
20110117	Salman	Noor	Khoper	M	018550695	snoor@hotmail.com
20110118	Hadi	Halwaji	Qatif	M	018350645	hhalwaji@hotmail.com
20110119	Noor	Jassim	Qatif	F	026550645	njassim@hotmail.com
20110110	Sultan	Rauf	Qatif	M	026650645	srauf@hotmail.com
20110129	Jawad	Muhammed	Qatif	M	026553345	jmuhammed@hotmail.com

LOGIN SUCCESSFUL:

You are logged in. Welcome admin1
[Logout](#)